SAMPLE

History of Astronomy: Indian Astronomy

Mathematical Skies

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Free sample: Introduction & Chapter 1

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Introduction

Somewhere in the first centuries of the common era, in a culture that measured time by the Moon and organized its skies into twenty-seven lunar mansions, a quiet revolution took place in prose. Scholar after scholar, writing in Sanskrit on palm-leaf manuscripts, worked out the mathematics of the heavens: how to predict eclipses, how to tabulate the positions of the planets, how to reconcile the wanderings of the Sun and Moon with the demands of ritual calendars. Their writings seeded a tradition that would run, unbroken and continually self-correcting, for well over a thousand years, produce some of the medieval world's most accurate astronomical tables, and lay mathematical groundwork that would eventually flow into the global story of science.

The subject also has a boundary worth drawing at the start, since the reader may arrive from adjacent interests: this is a book about the scientific tradition, the calendars, tables, instruments, and arguments, and not a survey of the subcontinent's cosmological myths, of astrology's interpretive rules, or of the many regional folk-astronomies, each of which has its own literature and deserves its own book. Where the scientific tradition touches those neighbors, the book says so, because the contacts are part of the science's history, but the through-line stays on the computation and the instruments, which is where the tradition earned its place in world science.

This is a book about that tradition: Indian astronomy, from its earliest traces in the Indus Valley and the Vedic hymns to the space telescopes of the twenty-first century. It is a subject that rewards patient telling, because the tradition is neither a footnote to Western astronomy nor a museum of curious beliefs, but a complete scientific culture with its own questions, methods, institutions, and arguments, one that ran in parallel with Greece, Babylon, China, and the Islamic world, exchanged ideas with all of them, and produced results, in certain eras, that no other civilization matched.

The story has moments that still startle readers who meet it for the first time. A scholar in the fifth century argued that the Earth rotates on its axis while the stars stand still. Another computed the circumference of the Earth with remarkable closeness, and explained eclipses as shadows rather than the doings of a devouring demon. A school in the far south of the subcontinent, centuries before Newton or Leibniz, developed infinite series for the sine and for the arctangent, stepping within a hand's breadth of calculus itself. All of this happened in Sanskrit, in verse, with rules for poetry governing the mathematics, and all of it happened in a living tradition whose practitioners wrote commentaries on their predecessors and corrected them openly.

A word on language and names, before the story starts. The tradition's texts are in Sanskrit, and its terms have entered English scholarship directly, so this book uses them with translations at first appearance: jyotisha for the discipline, siddhanta for the treatises, nakshatra for the lunar mansions, tithi for the Moon's day. Personal names are given in their standard scholarly romanizations, and dates are stated in the common era, with uncertainties flagged where the specialists flag them, since the tradition's own chronology uses several different eras, a puzzle the calendar chapter untangles.

The chapters ahead follow the plan the series keeps for every subject, bent where the tradition's own shape requires it. The discovery chapter says what the sources are and how the modern world recovered them. The understanding chapter explains what the discipline was, its genres, its training, and its place among the sciences of India. The characteristics chapter returns to the founding systems, the nakshatra belt and the lunisolar calendar, whose shapes the tradition kept for two thousand years. The science chapter walks the siddhanta mathematics, the observation chapter the instruments and the observatory culture, and the features chapter catalogs the tradition's signature parts, the genres, the verse, the zero, and the almanac.

The context chapter maps the exchanges with Greece, China, and the Islamic world. The modern-understanding chapter follows the colonial encounter, the tradition's recovery, and the calendar that never died, and the research chapter surveys the institutions, telescopes, and space missions of the republic. The discoveries chapter counts the tradition's significant results, from Aryabhata's arguments to the Kerala school's series, and the challenges chapter keeps the open questions honestly. A future chapter looks ahead, a trivia chapter shelves the pleasures, and the conclusion weighs the whole. Throughout, the book keeps its evidence labeled honestly, separating what the texts say, what the archaeology supports, and what remains debated among specialists, because a tradition this old deserves prose that careful.

A reader's note on scope's edges belongs here too: mathematics appears throughout this book, since the tradition's astronomy was inseparable from its algebra, but the book is an astronomy's history, and the reader seeking the tradition's mathematics in full, the indeterminate equations, the combinatorics, the famous decimal fractions debate, will find it treated in the companion volume on Indian mathematics, where it fills a book of its own.

The reader's own vantage deserves a sentence at the outset too: this tradition's evidence is textual in a way Greek astronomy's is not, so the reader will meet manuscripts, verses, and commentators throughout, and the book's habit of quoting the sources' reasoning in plain paraphrase is deliberate,

because the tradition's character, argumentative, cumulative, and courteous to its own classics, is part of the science itself, and a history that only tabulated results would miss the best part.

By the end, the reader should see Indian astronomy as what it was: one of humanity's longest continuous scientific conversations, conducted for two millennia in one of the world's great languages, about the oldest subject there is.

Chapter 1: The Discovery of Indian Astronomy

The discovery of this astronomy is therefore a discovery of its documents, since the tradition speaks to the modern reader only through what survived, and the survival is itself the story. The student of Indian astronomy is unusually blessed with sources, and the first task of any honest book is to say what they are and what they can and cannot tell us.

A brief orientation to the subcontinent's own deep background belongs here, flagged honestly as context rather than astronomy: the Indus Valley civilization, contemporary with the early Vedic age, built cities of striking geometric regularity whose weights and measures show advanced practical mathematics, but its script remains undeciphered, so any claim about its astronomy, calendar alignments above all, is inference from architecture rather than reading, and this book keeps such claims labeled as debated. What is certain is the substrate: a materially sophisticated, mathematically fluent urban culture, into whose successor societies the Vedic hymns arrived, and the meeting of the two, ritual sky-lore and practical mathematics, is the most plausible soil of the later tradition, with the emphasis on plausible.

The oldest layer is embedded rather than dedicated. The Vedas, the hymn collections composed in the second millennium before the common era, are not astronomy texts, but they name the stars and the months, describe a year of twelve lunar months with a periodic intercalary month, and set out the twenty-seven nakshatras, the lunar mansions along the Moon's path that organized all later Indian sky-lore. From these hymns and their ritual literature, scholars reconstructed a working lunisolar calendar of great antiquity, and the first text devoted entirely to astronomy, the Vedanga Jyotisha of roughly the last centuries before the common era, exists precisely to serve that calendar, giving rules for tracking the Moon and the solstices so that rituals could be performed on the right days. It is a short, practical text, and it shows astronomy being born as a servant of ritual timekeeping, an origin it shares with the calendars of Babylon and China.

The manuscript culture's own numbers deserve one paragraph, because they

are part of the tradition's achievement. The surviving Sanskrit astronomical corpus runs to many thousands of manuscripts across the subcontinent and beyond, from Nepal to the south, from temple libraries to European collections, with the major classical texts each surviving in dozens to hundreds of copies, an order of magnitude more than the Greek astronomical canon, and the copying tradition's accuracy, checked by collating independent manuscript lines, is high wherever it has been studied. The tradition, in other words, not only wrote its science; it copied and recopied it with a diligence that preserved the whole library to the present, which is a large part of why this book can exist.

The classical layer is the great one. Beginning around the early centuries of the common era, Sanskrit astronomy developed a standard genre, the siddhanta, meaning roughly "established conclusion": a comprehensive treatise that set out the cosmology, the mathematics, and the numerical tables needed to compute planetary positions, eclipses, and calendars. Dozens of these survive, along with a second genre, the karana, a shorter handbook of ready-made rules for practical computers, and a third, the commentary, in which later scholars explained, criticized, and improved their predecessors with a candor that modern historians treasure. The manuscripts themselves, written on palm leaf and later paper, survive in the thousands, copied and recopied by the tradition's working astronomers, and the great centers of Sanskrit learning kept the chain alive into the colonial period.

One class of inscription deserves the highlight, because it is the tradition's best external check: the carved and engraved records of eclipses and planetary meetings, made at temples and on royal grants across the subcontinent, sometimes as the event's commemoration, sometimes as the date's specification in astronomical terms. The modern astronomer can compute every such event backwards, and the checks, run repeatedly in the scholarly literature, confirm both the events and the inscribers' calendar competence at a high rate, with the mismatches themselves instructive, since they cluster where the regional calendar variants disagree, letting historians map the tradition's local practice as no other evidence can.

Around this literature stands a supporting archaeology: temple architecture aligned to the cardinal directions and to the rising points of particular stars; sundials and water clocks; inscriptions recording eclipses and planetary conjunctions, which modern astronomers can check backwards against computation, verifying both the events and the record-keepers' competence; and the huge observational installations of the eighteenth century, built in stone at a moment when the tradition met the telescope era. There is also the testimony of the neighboring civilizations, since Indian astronomy appears in

Greek, Chinese, Arabic, and Persian sources, and the cross-references, dates, and borrowings between traditions let historians anchor the Sanskrit chronology with unusual confidence.

The sources also have limits, stated plainly. A note on the tradition's own historiography belongs here, because it is a distinctive asset: the Sanskrit authors cite each other, name their teachers, and date their own works with the local era, and the resulting citation network, centuries of it, is dense enough that the classical tradition's internal chronology is better anchored than most pre-modern sciences anywhere. When a date in this book is stated confidently, the citation network is usually the reason; when it is hedged, the text in question stands outside the network's reach, and the hedge is the scholar's, not the author's.

Dates for the earliest figures rest on internal evidence from the texts, cross-tradition references, and astronomical configurations embedded in the works, and they carry genuine uncertainties, as later chapters note. Some major works are known only through later quotations because the originals are lost. And until the modern era of critical editions and translations, access ran through a thin channel of European scholarship, which arrived with its own agenda and errors, as the chapters on the colonial period describe. But the balance sheet is strong: a continuous textual tradition of two thousand years, checked against eclipses, inscriptions, and neighbors' records, gives the history of Indian astronomy an evidential depth surpassed by only one or two other scientific traditions on Earth.

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